The Economic Origins of the Constitution Revisited: A Novel Analysis using Fuzzy String Matching

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Charles Beard's seminary work, *An Economic Interpretation of the United States Constitution,* contends that the Constitutional Convention was attended by, and the Constitution was therefore written by, a "cohesive" elite seeking to protect its personal property (especially Continental bonds) and economic standing. With the benefit of more detailed data, Forrest McDonald (1958) revisited this question. In contrast to Beard, he concluded that most of the delegates to the Constitutional Convention and the various state conventions were motivated by what they viewed as best for their individual states and the federation of all 13 states rather than their personal financial interests. Our goal in this study is to revisit this question using datasets containing records of the four main types of debt created by the Revolutionary War. Our goal here is primarily descriptive - we aim to describe the various assets of delegates and present measures that depict differences among the delegates based off their final votes.

Introduction

An Economic Interpretation of the Constitution, published in 1913, describes the Constitution as a mechanism for wealthy men who held bonds, which were considered personal property, rather than just land, classified as real property, to subdue tendencies created by the Revolution held by common people, including their debtors. In guaranteeing the Constitution and as such, the establishment of a nation, they enabled a path for their debt to be transferred into something tangible and in doing so, weakening those who owed them money. This interpretation of classism began to lose ground in the 1950s, and in 1958, Forrest McDonald revisited this question in *We the People: Economic Origins of the Constitution.* In contrast, McDonald concluded that the supposed two-class conflict was much more complicated than what Beard had described, most of the delegates to the Constitutional Convention and the various state conventions were motivated by what they viewed as best for their individual states and the federation of all 13 states rather than their personal financial interests. Our goal is to revisit this subject and utilize our new datasets on loans and debts from the American Revolutionary War and the post-revolutionary era to determine the economic incentives of delegates who were involved within the ratification process of the constitution. The analysis contained within this paper is primarily data-driven and focuses less on historical context and issues such as classism, and more on an economic perspective - what the numbers, a delegate's wealth, informs us. Using the advanced computational tools available at our disposal, we use

fuzzy string matching to identify the owners of debt certificates, allowing us to account for differences in the certificates such as the presence or absence of middle names or variations in spelling.

Literature Review

Fuzzy matching has been utilized in a variety of research studies and its usage is particularly well documented in economics. For example, Andrade et al. (2018) investigates the impact of entrepreneurship on cooperative job creation in a large North American post-secondary institution. The authors achieve this by running fuzzy matching on two datasets: a list of companies started by the institution's engineering students and faculty and a database of the institution's cooperative education (co-op) placements.

In addition, Dong (2020) examines the use of twitter in predicting bond trading activity. The author completes this by downloading all tweets with the query "$SPY" to find all twitter posts that coincide with SPY, the the largest Exchange-Traded Fund that tracks the S&P500. However, there were many duplicate tweets in the data due to their similarity. To solve for this, Dong paired up every tweet in the data with each other. Then, using fuzzy matching, all pairs with the Levenshtein ratio above 90 were located, allowing for the author to eliminate the newer of the pair.

What separates our paper from both Beard and McDonald's work, as well as the work of other papers who have used fuzzy string matching is that we are hoping to apply this concept towards economic history. This is a field for which this technique has great potential due to the prevalence of historical non-numeric data as keys, which can often be messy and unsuitable for analyses that only use other techniques like exact matching or regular expressions.

Data

Pandas DataFrames constructed from the delegates from the 1787 Philadelphia Constitutional Convention and delegates to the state constitutional ratification conventions formed the comparison set.

Data was analysed in two categories: pre-1790 debt certificates and post-1790 securities. The available pre-1790s data consisted of: loan office certificate debt, state liquidated debt, Pierce certificates and marine liquidated debt. Although the records are incomplete, they represent approximately 85% of domestic debt. DataFrames were constructed for each category to be fuzzy matched to the delegates.

The post-1790s data consisted of state loan office ledgers recording issues of three different types of stocks: six per cent, deferred six per cent, and three per cent. These stocks were issued as per Alexander Hamilton's refinancing plan. Records from Massachusetts and Delaware are entirely missing, while some other states are incomplete. When available, data from each state was written into separate DataFrames for assumed state debt and continental debt.

Before commencing *fuzzy wuzzy* string matching, the above data was cleaned. These are the special cases that were considered and how they were notated:

1) Companies or multiple people in one entry: if given two full names, were separated into full name 1 and full name 2 columns. Otherwise, they were sorted into a list for further research.
    a) Ex: 'Peter & Anna Maricha' cleaned to 'Peter Maricha' and 'Anna Maricha'
    b) Ex: 'Clark & Nightingale' most likely corresponds to John Clark and Joseph Nightingale

2) Additional details in an entry: stripped from the name and added into a separate notes document.
    a) Ex: ‘Capt Samuel Wilman and Co.’ cleaned to ‘Samuel Wilman’ with the title ‘Capt’ and the note ‘Co.’
    b) Ex: ‘Moses Bush & Sons’ cleaned to ‘Moses Bush’ with the note ‘Sons’
3) Executors to individuals or estates: separated into two full names with the executor labeled as such
    a) Ex: ‘Joseph Allen Executor to B Winchester’ cleaned to ‘B Winchester’ and ‘Joseph Allen’ with the title ‘executor’

Methods

The cleaned data was run through a 3-step fuzzy matching algorithm to determine the holdings of each delegate. We devised this algorithm knowing that Levenshtein distance, which is the basis for how fuzzy matching scores are generally calculated, was effective informing us whether there were minor differences in a name’s spelling (such as the difference between John and Johne) but not for other issues such as the presence or absence of middle name(s). Our two input lists were the list of delegate names and the list of names from the debt certificates.

Our first step used the process.extract function to produce a list of potential matches from these two lists, which we defined as those having passed a threshold score of 85. Process.extract combines the four different types of ratios in fuzzywuzzy into one score.

For our second step, define a matching pair to be a match from the list of matches we produced in step one. For each matching pair, which has two names, let m = # of unique words in name 1, and let n = # of unique words in name 2, and let M and N equal the set of unique words in name 1 and 2, respectively. We only compared matches where both m and n were greater than or equal to two; all other matches were discarded. Let the threshold number be defined to be min(m, n). For each possible pairing of words m_i in M and n_i in N, we want to keep all matches where the number of pairings that have a process.extract score of above 90. Name 1 and Name 2’s assignment are arbitrary as long as all M and all N come from distinct sets (as in all M either belong to all delaget names with a match or all debt names with a match).

In the third step, we examine whether a particular debt name has been matched to multiple delegates. If so, we remove all matches including that particular debt name except the one with the highest original process.extract score. We also conduct some additional manual cleaning based off the eye test, which you can see in our publicly available code. Our parameters for the scores were set based off of our observations of what seemed to reasonable capture names that could be considered as similar.

The algorithm aimed to set a lower bound by only yielding certain matches. Although some ambiguous cases were missed, the matches reported indicate a minimum amount of debt owned. The match list and loan certificate data were then merged together to determine the total specie value and certificate count for each matched delegate.

Since we only matched names, cases of different people with the same common name were evaluated as one person. For example, John Smith has many assets because we could not differentiate between

separate people. Some ways to mitigate this problem might be to match the state the debt came from or match titles if given.

Results/Discussion

After matching delegates to their asset totals, the first thing we wanted to do was to learn more about what the data contained.

This is a table of the top 10 delegates who held the most debt, which we defined to be the total value of the face value of their loan office certificates holdings and their state certificates. Our rationale for this definition of debt was that we assumed state certificates were not tied to specie value, so it would be more appropriate to aggregate them to the loan office certificates' face value, not the specie value. Specie value is the value of a debt certificate tied to some form of hard currency, like gold, while face value was the value of the debt in whatever currency it was going to be paid out in. The Pierce Certificates were backpay for Continental soldiers and therefore were more spread out than the loan office or state certificates. Additionally, we had concerns about the accuracy of name matching due to common names (see the Pierce table for more details).

In addition, Pierce Certificates and Marine certificates were backpay, which we considered to be different from actual debt. This was because debt was bought, while the two types of certificates mentioned here was simply money that they were owed, not money that they had paid money to be owed. As such these were two different classes of assets from the human perspective. We also believed that because records of pay were more difficult for individuals thus, delegates would have only been able to realistically calculate their assets that consisted of loan office and state certificates, if they were using money to inform their decisions, so as such, we only chose those two assets to form our definition of total debt assets.

Table 1: Wealthiest Delegates Ordered by Total Debt Assets

First Name	Last Name	Delegate's state	Vote	Loan Office Certificates Total Face Value	State Certificates Total Dollars	State Certificates 90th	Total Debt Assets
Robert	Morris	Constitutional convention	yes	405700	1053	37	406790
John	Stevens	NJ	yes	399300	3463	49	402812
Jeremiah	Wadsworth	CT	yes	303300	22	45	303367
William	Smith	SC	yes	212200	5927	49	218176
Robert	Coleman	PA	yes	206200	0	0	206200
John	Skinner	NC	yes	203400	15	45	203460
John	Davis	MA	yes	134900	2171	62	137133
Davis	John	RI	yes	134200	2171	62	136433
John	Dunlap	MA	yes	112100	1135	74	113309
Williams	John	RI	no	107800	1772	39	109611

These results align with our present knowledge of the wealth of delegates - for example, we knew that Robert Morris was one of the wealthiest men in colonial America, and among all constitutional and state convention delegates, he stands at the top.

Now, we present a more detailed breakdown of their holdings, by asset class. This is a table of the top 10 delegates who held the most debt, in terms of the value of just their loan office certificates' face value.

Table 2: Wealthiest Delegates Ordered by Loan Office Certificates Assets (Face Value)

First Name	Last Name	Delegate's state	Vote	Loan Office Certificates Total Face Value	Loan Office Certificate Count
Robert	Morris	Constitutional Convention	yes	405700	309
John	Stevens	NJ	yes	399300	344
Jeremiah	Wadsworth	CT	yes	303300	369
William	Smith	SC	yes	212200	381
Robert	Coleman	PA	yes	206200	306
John	Skinner	NC	yes	203400	313
John	Davis	MA	yes	134900	196
Davis	John	RI	yes	134200	194
John	Dunlap	MA	yes	112100	72
Williams	John	RI	no	107800	176

Here is the same table, for their certificates' specie value.

Table 3: Wealthiest Delegates Ordered by Loan Office Certificates Assets (Specie Value)

First Name	Last Name	Delegate's state	Vote	Loan Office Certificates Total Specie Value	Loan Office Certificate Count
Robert	Morris	Constitutional convention	yes	67275.1673	309
William	Phillips	MA	yes	44598.9388	170
Jeremiah	Wadsworth	CT	yes	35498.5298	369
William	Smith	SC	yes	29479.0972	381
John	Stevens	NJ	yes	27727.4954	344
John	Langdon	Constitutional convention, NH	yes	26977.6117	112
Brown	John	RI	yes	26084.5709	58
John	Brown	NC	no	26084.5709	58
William	Moore	DE	yes	24908.0486	90
Samuel	Griffin	NH	yes	24838.7113	41

Here is the same table, for the value of the state certificates.

Table 4: Wealthiest Delegates Ordered by State Certificates Assets

First Name	Last Name	Delegate's state	Vote	State Certificates Total Dollars	State Certificate Count
William	Elliot	SC	yes	57320	5
David	Deshler	PA	yes	15581	11
Robert R.	Livingston	NY	yes	10721	6
William	Butler	SC	no	10622	1
Elbridge	Gerry	Constitutional convention	no	10100	1
James	Thompson	NC	no	8486	18
James	Morris	PA	yes	8163	4
William	Brown	PA	no	7957	132
Thomas	Gaskins	VA	yes	7200	1
John Jacob	Faesch	NJ	yes	6273	7

Here is the same table, for the value of the Pierce certificates.

Table 5: Wealthiest Delegates Ordered by Pierce Certificates Assets

First Name	Last Name	Delegate's state	Vote	Pierce Certificates Total Value	Pierce Certificate Count
John	Smith	NY	yes	29545.9	241
John K.	Smith	MA	yes	25384.62	207
William	Scott	SC	yes	15801.36	50
William	Moultrie	SC	yes	14423.6	6
William	Smith	SC	yes	14004.65	102
William	Heath	MA	yes	13436.74	10
Anthony	Wayne	PA	yes	11902.63	8
Davis	John	RI	yes	11639.91	69
John	Davis	MA	yes	11639.91	69
Samuel H.	Parsons	CT	yes	10840.64	14

Notice that the top two debt holders are John Smith, with certificate counts over 200. This indicates that we have aggregated debt from many different people under one name; compare John Smith or William Smith to William Moultrie, who only owns 6 certificates. For this reason, we have omitted the Pierce certificate data from our total debt assets calculations.

Here is the same table, for the value of the Marine certificates.

Table 6: Wealthiest Delegates Ordered by Marine Certificates Assets

First Name	Last Name	Delegate's state	Vote	Marine Certificates Total	Marine Certificate Count

				Dollars	
William	Jones	MA	no	1872	5
George	Houser	NC	yes	1047	1
William	Smith	SC	yes	636	1
William	Morris	MD	yes	471	3
John	Hunter	SC	yes	310	1
John	Brown	NC	no	262	1
Brown	John	RI	yes	262	1
John	Elliot	GA	yes	194	1
Benjamin	Browne	MA	yes	141	1
David	Bell	VA	no	108	1

We assume that the problem of the Pierce certificates does not apply to the Marine certificates due to the low certificate counts. However, we found a total of 31 matches to the Marine certificates and are unsure how much the small value of the certificates contribute to a delegate's total wealth.

Next, we decided to find the top 10 richest delegates who voted yes or no, again decided by the value of their total debt assets. Here's the table for delegates who voted no.

Table 7: Wealthiest Delegates Who Voted No Ordered by Total Debt Assets

First Name	Last Name	Delegate's state	Vote	Loan Office Certificates Total Face Value	State Certificates Total Dollars	State Certificates 90th	Total Debt Assets
Williams	John	RI	no	107800	1772	39	109611
James	Thompson	NC	no	96400	8486	40	104926
Elbridge	Gerry	Constitutional convention	no	76500	10100	0	86600
Thomas	Brown	NC	no	74500	1209	9	75718
John	Hamilton	MA	no	55300	103	3	55406
William	Jones	MA	no	43100	1161	18	44279
William	Thompson	MA	no	39800	1364	12	41176
John	Brown	NC	no	35600	1208	19	36827
Thomas	Chase	NH	no	30500	1440	0	31940
John	Taylor	MA	no	28500	3109	50	31659

Here's the table for delegates who voted yes.

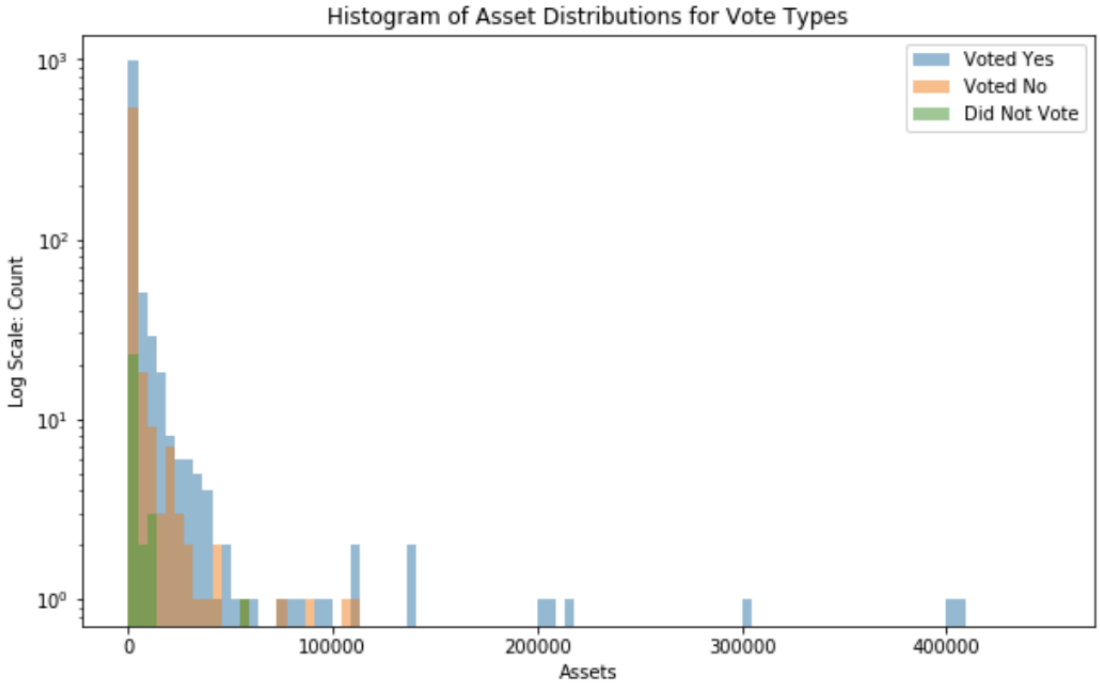
Table 8: Wealthiest Delegates Who Voted Yes Ordered by Total Debt Assets

First Name	Last Name	Delegate's state	Vote	Loan Office Certificates Total	State Certificates Total Dollars	State Certificates	Total Debt

				Face Value		90th	Assets
Robert	Morris	Constitutional convention	yes	405700	1053	37	406790
John	Stevens	NJ	yes	399300	3463	49	402812
Jeremiah	Wadsworth	CT	yes	303300	22	45	303367
William	Smith	SC	yes	212200	5927	49	218176
Robert	Coleman	PA	yes	206200	0	0	206200
John	Skinner	NC	yes	203400	15	45	203460
John	Davis	MA	yes	134900	2171	62	137133
Davis	John	RI	yes	134200	2171	62	136433
John	Dunlap	MA	yes	112100	1135	74	113309
John	Williams	VA	yes	107500	1772	39	109311

Here's a better visual depiction of the contrast in assets between delegates who voted differently. Although delegates who voted yes were slightly more likely to have larger assets, as depicted in the histogram, the majority of delegates who were in any of the categories had either no, or little assets.

Figure 1: Histogram of Asset Distributions for Vote Types



The next thing we did was to explore the vote distributions of the delegates who voted yes and no. We wanted to see whether there was a difference, just because a fair number of the delegates could not be matched to any debt assets at all. Here is a chart of the delegates with and without assets.

Figure 2: Count Distribution of Delegate Votes by Asset Possession

```
Delegates with 0 debt assets
yes             677
no              394
did not vote     16
Name: (Identifier, sign), dtype: int64
Delegates with debt assets
yes             447
no              197
did not vote     13
Name: (Identifier, sign), dtype: int64
```

Here's the same table, but this time weighted with proportions.

Figure 3: Proportion Distribution of Delegate Votes by Asset Possession

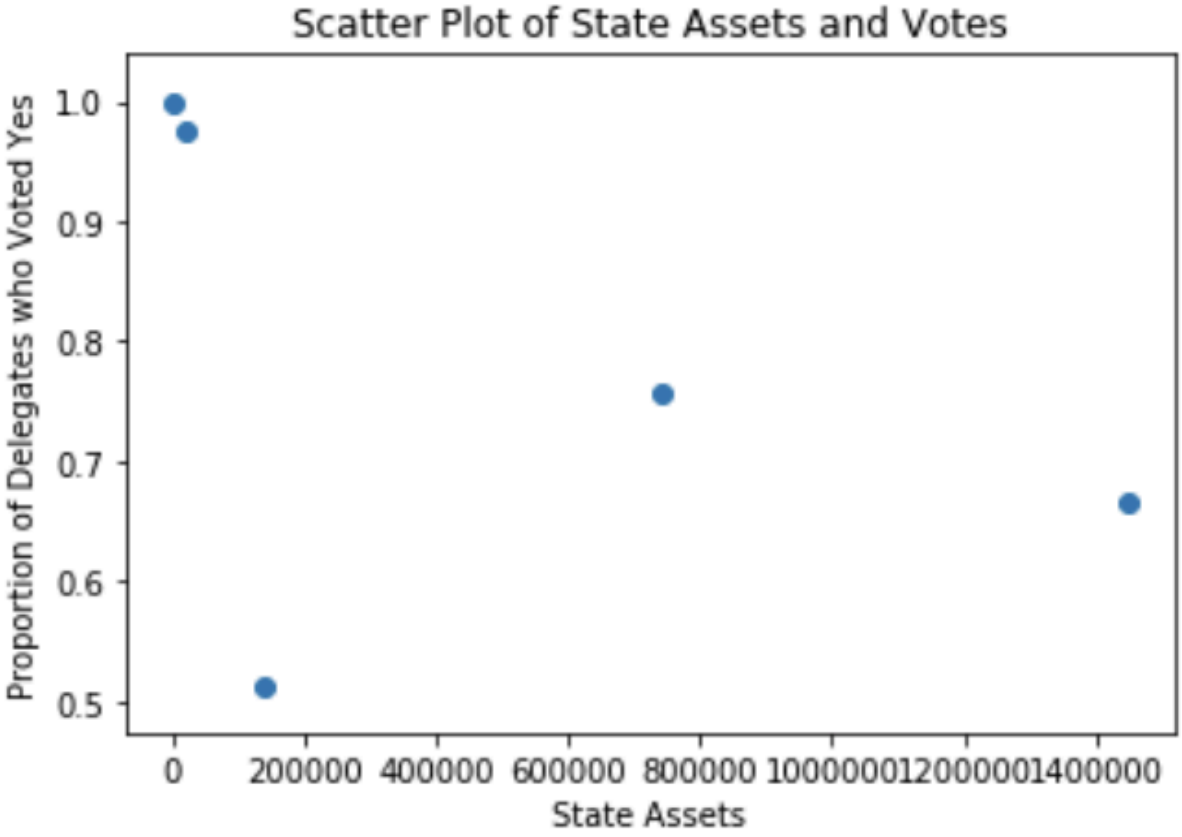
```
Delegates with 0 debt assets percentages
yes             0.622815
no              0.362466
did not vote    0.014719
Name: (Identifier, sign), dtype: float64
Delegates with debt assets
yes             0.680365
no              0.299848
did not vote    0.019787
Name: (Identifier, sign), dtype: float64
```

What we see is that there's a difference between the proportion of delegates who voted yes and no depending on whether they had assets or not, a difference of around 6%. This seems to suggest that although it was probably not the deciding factor that helped motivate a delegate's decision, the presence or absence of assets certainly was a factor that affected a delegate's vote. The proportion of delegates who did not vote did not change much.

We next explored whether there was any relationship between a state's total debt holdings and the likelihood its delegates voted yes or no for the constitution. To do this we aggregated their debt under one category called total using the post1790 debt certificates. We found that it seemed like there was a general downward trend towards having less delegates from a state vote yes for the constitution if the state had more debt, but the paucity of data available (we had 5 data points) meant that this conclusion was likely of no significance. However, it is important to note that our finding contradicts that of Forrest McDonald, who claimed delegates voted in their state's interest, since it appears that the more debt a state was owed, the lower the proportion of delegates from that state who voted no. Here is a chart of the scatter plot.

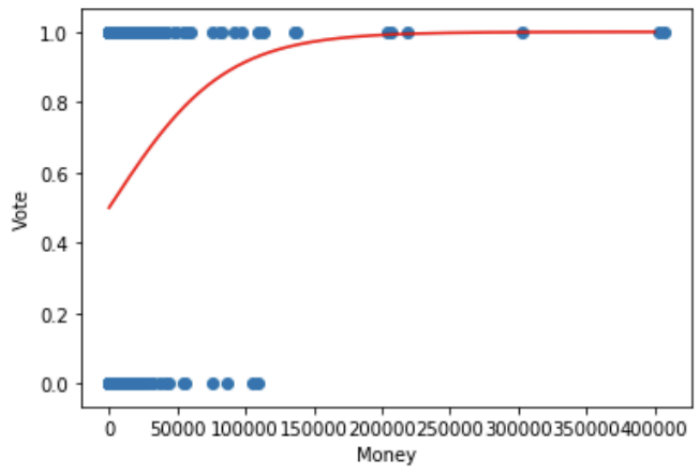
Figure 4: Scatter Plot of State Debt and Proportion of Delegates who Voted Yes

Points from left to right are CT, GA, NH, NJ, SC



Finally, we ran a logistic regression to see whether there was evidence that increased assets were correlated with an increased tendency to vote yes. What we found was that a lot of the delegates who were extremely wealthy voted yes, but there were also plenty of wealthy delegates who voted either way, and many delegates without assets who also voted either way. Our regression was significant, but a visual test shows that there's no clear boundary demonstrating evidence that having more wealth is a good predictor of whether someone will vote yes. Here is our regression summary and chart.

```
Vote = Debt_Assets*2.3720207068226288e-05
Vote is encoded as a binary variable, 1 = yes and 0 is anything else (no, didn't vote)
                           Logit Regression Results
==============================================================================
Dep. Variable:     ('Identifier', 'sign')   No. Observations:                 1744
Model:                              Logit   Df Residuals:                     1743
Method:                               MLE   Df Model:                            0
Date:                    Sun, 15 Aug 2021   Pseudo R-squ.:                -0.05291
Time:                            16:24:14   Log-Likelihood:               -1195.0
converged:                           True   LL-Null:                      -1135.0
Covariance Type:                nonrobust   LLR p-value:                      nan
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
x1           2.372e-05   6.22e-06      3.812      0.000    1.15e-05    3.59e-05
==============================================================================
```



Our conclusion is that there is no good evidence that more assets meant increased likelihood to vote yes given how poorly the regression fit, but there is evidence to suggest that not having assets meant greater likelihood of not voting yes.

Future Work

Here are some brief notes on directions this project can take in the future.

1) Fuzzy Matching
    a) Improve the fuzzy matching by separating out company names into the names of their owners
        i) This could also segway into some pretty interesting network analysis
    b) Take into account the titles and properties of individuals, such as whether they were an executor, and whether money was for an individual or estate
    c) Encode for the abbreviations of names (Jn = John) to make it easier for the matching algorithm to identify them
    d)
1) McGuire Data
    a) We have a detailed dataset on the assets and other information about the delegates to the constitutional convention, and we could incorporate this into helping us figure out the motivations behind their vote

2) State vs Constitutional Convention
    a) Perform more detailed analysis distinguishing the two types of conventions
3) Post-1790 Certificates
    a) Analyze the individuals who received 1790 debt payments, compare to the individuals who owned pre 1790 debt

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